

RESEARCH ARTICLE

10.1002/2014WR016828

Key Points:

- Developed a stochastic programming with recourse model hydropower production
- Performed in-sample and out-sample tests to evaluate scenario reduction impact
- Evaluated the tradeoff between computational demand and scenario reduction level

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Citation:

Xu, B., P.-A. Zhong, R. C. Zambon, Y. Zhao, and W. W.-G. Yeh (2015), Scenario tree reduction in stochastic programming with recourse for hydropower operations, *Water Resour. Res.*, 51, 6359–6380, doi:10.1002/2014WR016828.

Received 20 DEC 2014

Accepted 13 JUL 2015

Accepted article online 21 JUL 2015

Published online 16 AUG 2015

Corrected 27 APR 2016

This article was corrected on 27 APR 2016. See the end of the full text for details.

Scenario tree reduction in stochastic programming with recourse for hydropower operations

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Abstract A stochastic programming with recourse model requires the consequences of recourse actions be modeled for all possible realizations of the stochastic variables. Continuous stochastic variables are approximated by scenario trees. This paper evaluates the impact of scenario tree reduction on model performance for hydropower operations and suggests procedures to determine the optimal level of scenario tree reduction. We first establish a stochastic programming model for the optimal operation of a cascaded system of reservoirs for hydropower production. We then use the neural gas method to generate scenario trees and employ a Monte Carlo method to systematically reduce the scenario trees. We conduct in-sample and out-of-sample tests to evaluate the impact of scenario tree reduction on the objective function of the hydropower optimization model. We then apply a statistical hypothesis test to determine the significance of the impact due to scenario tree reduction. We develop a stochastic programming with recourse model and apply it to real-time operation for hydropower production to determine the loss in solution accuracy due to scenario tree reduction. We apply the proposed methodology to the Qingjiang cascade system of reservoirs in China. The results show: (1) the neural gas method preserves the mean value of the original streamflow series but introduces bias to variance, cross variance, and lag-one covariance due to information loss when the original tree is systematically reduced; (2) reducing the scenario number by as much as 40% results in insignificant change in the objective function and solution quality, but significantly reduces computational demand.

1. Introduction

In hydropower management and operation, decision makers are required to schedule releases so that overall benefit can be maximized. This requires the determination of the releases for the immediate time period, facing uncertain future inflows. Traditionally, streamflow forecasts have been utilized to assist in decision making. However, since long-term (monthly, seasonally, etc.) meteorological forecasts are unreliable, it is difficult to predict streamflow accurately on a long-term scale [Mao *et al.*, 2000; Tucci *et al.*, 2003]. This is because the meteorological factors as well as land surface [Samaniego and Bardossy, 2006] and soil [Hossain *et al.*, 2004] information that determine the streamflow have stochastic [Li *et al.*, 2009; Zhao *et al.*, 2011] and chaotic properties [Sivakumar *et al.*, 2001a, 2001b; Moriasi *et al.*, 2007]. Since long-term streamflow forecasts are highly uncertain, the optimal operation of reservoirs is basically a risk-based decision-making problem.

Multistage stochastic programming with recourse models [Birge and Louveaux, 2011; Yeh, 1985] can be used to assist in making release decisions under uncertainty. In stochastic programming, a streamflow series is considered to be a stochastic process. The original continuous streamflow series is discretized and represented by a limited number of discretized scenarios. Then the stochastic programming model uses streamflow scenario trees to represent random streamflow outcomes that may occur in the future. Consequently, the original stochastic model can be converted to a deterministic equivalent formulation. The streamflow scenario trees can be extracted from the historical streamflow series and serve as inputs to the stochastic programming with recourse model. In general, scenario tree generation methods [Gulpinar *et al.*, 2004; Hayland and Wallace, 2001] include simulation-based methods, optimization methods, and clustering-based

methods. The simulation-based methods [Trezos and Yeh, 1987; Kaut and Wallace, 2007; Turgeon, 2005] sample the scenarios using the distribution of random variables and preserve the observed transition probability. However, it is difficult for the simulation-based methods to sample scenario trees from multivariate distributions because cross correlations exist. The moment matching methods, which attempt to match the statistical moments (mean value, variance, skewness, kurtosis, and covariance) of scenario trees with the statistical moments of observed samples, use optimization algorithms to find the best match. Dupacova et al. [2000] compared different methods and criteria in scenario tree generation and suggested that the tree generation method should be selected according to the specific problem under consideration. Høyland and Wallace [2001] developed an algorithm that minimizes the differences between statistical moments of scenario trees and observed samples. This method allows mismatch between the moments of scenario trees and observed samples, since some of the moments are extremely difficult to match. Under this situation, decision makers have to compare the importance of each type of moment and set proper weights for minimizing the weighted sum of the difference between the statistical moments of scenario trees and observed samples.

Clustering-based methods offer an alternative way to construct scenario trees. Instead of using the distribution of the random variables, they sample directly from observed samples and find the representative scenarios by clustering. As a result of the clustering procedures, the generated scenario tree usually includes the scenarios that represent the clustering centroids of the observed samples; thus the mean value of the scenario tree usually matches that of the observed samples. Heitsch and Römisch [2009] proposed a method that uses the forward and backward reduction and a bundle algorithm to generate a scenario tree according to some heuristic rules. Latorre et al. [2007] compared four typical clustering algorithms to generate scenario trees, and found that the neural gas method outperforms the other three methods. The neural gas method [Martinetz et al., 1993] is a machine learning algorithm and is renowned for vector quantization from observed samples. Unlike the other clustering-based methods proposed by Casey and Sen [2005] and Heitsch and Römisch [2009] that use scenarios combining and deleting techniques to change the prespecified tree structure during the clustering process, the neural gas method preserves the prespecified tree structure at all times. This method approaches the centroids of the observed samples through an iterative update of the value of the nodes in the scenario tree in such a manner that the distance from the tree to the observed samples is reduced gradually. Due to its superior performance, we use this method to generate scenario trees in this study.

A stochastic programming with recourse model suffers from a heavy computational burden [Pan et al., 2015] as the dimensionality of the scenario tree grows. The computational demand of the model is directly related to the size of the scenario tree. Therefore, scenario tree reduction can be an effective way of decreasing the computational burden and has received much deserved attention [Growe-Kuska et al., 2003; Heitsch and Römisch, 2007]. However, scenario tree reduction causes information loss, since fewer scenarios are used to capture the characteristics of the observed samples. Consequently, the result obtained from a reduced model (a model based on a reduced tree) may be suboptimal due to the information loss. On one hand, we wish to reduce the computational demand as much as possible. On the other hand, we also wish to preserve solution accuracy when scenarios are reduced. Since the number of scenarios determines both the computational demand and solution accuracy [Høyland and Wallace, 2001; King and Wallace, 2012], analyzing the trade-off between the computational demand (CPU time) and solution accuracy is very important.

Most scenario tree reduction techniques reduce similar scenarios in which the distance between them is close, but without considering the impact on solution accuracy caused by scenario tree reduction. Dupacova et al. [2003] proposed an algorithm that determines the optimal subset of scenarios having probability metrics closest to the full set. De Oliveira et al. [2010] developed a method for reducing the scenarios that have the smallest Fortet-Mourier distance, and applied a stage-wise backward reduction. Da Costa et al. [2006] proposed a scenario tree reduction method using principal component analysis. It is worth noting that most studies have focused on how to reduce the scenario tree using the information based on the tree itself, rather than the consequences of scenario tree reduction on solution accuracy. In a stochastic programming with recourse model, it is more important to evaluate the impact of scenario tree reduction on the solution accuracy of the model. Instead of combining close scenarios, Housh et al. [2013] constructed a method by aggregating the scenarios that produce similar solutions. Specifically, this method individually

separates each scenario from the tree and uses a deterministic model to obtain the solution for each scenario independently. It then measures the distance between the solutions and identifies those solutions that are similar. Although this method associates the scenario with its corresponding solution using a one-to-one match, it does not account for the joint impact of scenarios on the solution. *Follestad et al.* [2011] analyzed the impact of scenario tree reduction on the objective function in a single reservoir system, but did not examine the impact of scenario tree reduction on the solution.

In this study, we identify the possible impact of scenario tree reduction on the solution accuracy of a stochastic programming model for hydropower operations. This helps determine the fewest number of scenarios that can be used in the stochastic programming model without introducing significant bias to the solution's accuracy. First, we conduct numerical experiments by systematically reducing the scenarios. For each reduced scenario tree, we run a stochastic programming model (without recourse) to determine the objective function and record the CPU time required for each model run. We use the in-sample and out-of-sample tests suggested by *Kaut and Wallace* [2007] to evaluate the impact of scenario tree reduction on the objective function of the stochastic model. We then assess the quality of the reduced solution (the solution based on a reduced tree) in terms of its energy production when this solution is implemented in real-time operations. Specifically, we select three typical hydrology years (a wet, normal, and dry year) to represent three different inflow patterns that may occur in the future and run a stochastic programming with recourse model for each reduced scenario tree under each inflow pattern. This study is distinct from all previous studies [*Follestad et al.*, 2011; *Kracman et al.*, 2006; *Lee et al.*, 2008; *Li et al.*, 2006; *Seifi and Hipel*, 2001; *Zambon et al.*, 2012; *Faber and Stedinger*, 2001; *Watkins et al.*, 2000; *Watkins et al.*, 2000; *Jacobs et al.*, 1995] in that we use nonlinear programming to solve the deterministic equivalents of the stochastic programming with recourse model for hydropower optimization. Nonlinear programming demands much more computational time than linear programming. Therefore, it is important to reduce the size of the nonlinear programming problem. This can be achieved by reducing the size of the scenario tree, but at the expense of degrading solution accuracy. Traditional tree reduction techniques reduce scenarios only considering the tree information, disregarding the influence of reduction on solution accuracy. In this study, we propose a tree reduction methodology that is based on the influence of reduction on solution accuracy. Specifically, we determine the trade-off between the level of tree reduction and the accuracy of the model solution associated with each level of tree reduction. The trade-off relationship allows decision makers to make an informed decision with regard to the appropriate level of tree reduction.

This paper is organized as follows. Section 2.1 outlines the neural gas method for generating a streamflow scenario tree. Section 2.2 formulates the objective function and constraints of the stochastic programming with recourse model for hydropower optimization. Section 2.3 outlines the procedures for the numerical experiments. We then introduce the in-sample and out-of-sample tests for assessing the impact on the objective function caused by scenario tree reduction in section 2.3.1 and present the procedures for solution quality assessment in section 2.3.2. Section 3 applies the proposed methodology to the Qingjiang cascade system of reservoirs in China and conducts the numerical experiments.

2. Methodology

2.1. The Neural Gas Method for Scenario Tree Generation

A streamflow series is considered to be a continuous stochastic process. A streamflow scenario tree is a discretized representation of the stochastic process. It consists of scenarios that can be extracted from observed samples, distribution functions, or expert options.

We define a streamflow scenario tree as $\{\omega_i\}$, $i=1, 2, \dots, I$ and set its structure, in which I is the total number of scenarios. Scenario ω_i consists of a sequence of nodes $\omega_{i,t}$, starting from time period one and ending at time period T , in which T is the total number of time periods in the planning horizon. Specifically, if scenario ω_{i_1} shares the same node $\omega_{i_1,t}$ with scenario ω_{i_2} in time period t , then $\omega_{i_1,t} = \omega_{i_2,t}$. Each node $\omega_{i,t}$ contains a streamflow vector $(\omega_{i,t,1}, \omega_{i,t,2}, \dots, \omega_{i,t,m}, \dots, \omega_{i,t,M})$ that represents the unregulated streamflow into the reservoir system during time period t , in which $\omega_{i,t,m}$ is the unregulated streamflow into reservoir m in scenario i and time period t (m^3/s), and M is the total number of reservoirs. Figure 1 shows the structure of a streamflow scenario tree. Note that the nodes that are shared by more than one scenario can be replaced by their equivalent nodes. For example, node $\omega_{5,2}$ can also be replaced by $\omega_{6,2}$ in the figure.

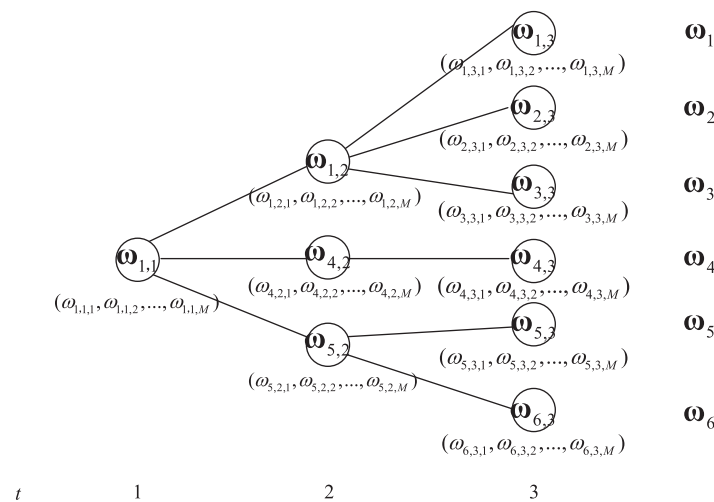


Figure 1. Streamflow scenario tree structure.

vector quantization process by randomizing the coordinates of vectors, and changes their positions for reducing the overall distance to the observed vector samples through gradual adaptations until the overall distance is minimized.

The following steps are involved:

1. Initialization

For each node in the tree, we initialize its value by selecting an observation record from the historical streamflow series at random:

$$\omega_{i,t} = \mathbf{IN}_{rand(\cdot),t}, \quad i \in [1, I], t \in [1, T], \quad (1)$$

in which $\mathbf{IN}$ is the historical streamflow series that includes the observation records for unregulated streamflow into each reservoir (m^3/s); $rand(\cdot)$ is a uniformly distributed random number in the range (1, K); and K is the total number of yearly historical streamflow series.

2. New series selection and distance-order calculation

Before each iteration, an entire series $\mathbf{IN}_k$ is randomly selected from $\mathbf{IN}$. We then evaluate the distance from each scenario to the selected series using the following equation:

$$d_{i,k} = \|\omega_i - \mathbf{IN}_k\|, \quad i \in [1, I], \quad (2)$$

in which $d_{i,k}$ is the Euclidean distance from scenario i to series k . We sort the distance array d in ascending order and record the order sequences in array O .

3. Iteration

To reduce the distance between the scenario tree and the selected series, we alter the position of the featured vectors (scenario tree) in order to move them toward the selected vector (selected series). This is accomplished by iterating on node values. The value of each node is iterated according to its scenario distance orders and the iteration times using the following equation:

$$\Delta \omega_{i,t} = \varepsilon(j) \cdot \sum_{i'=1,2,\dots,I|\omega_{i',t} \in \omega_{i,t}} h(O_{i'}, \lambda(j)) \cdot (\mathbf{IN}_k - \omega_{i',t}) / \sum_{i'=1,2,\dots,I|\omega_{i',t} \in \omega_{i,t}} 1, j \in [1, jm], \quad (3)$$

in which $\varepsilon(j) = \varepsilon_0 \cdot (\varepsilon_f / \varepsilon_0)^{j/jm}$ is the step size that reduces from ε_0 to ε_f as iteration time j is increased from 0 to j_m ;

$h(O_{i'}, \lambda(j)) = \exp(-O_{i'} / \lambda(j))$ is an exponential function that determines the adaptation value of the given node regarding its scenario distance order; and $\lambda(j) = \lambda_0 \cdot (\lambda_f / \lambda_0)^{j/jm}$ changes from λ_0 to λ_f as the iteration continues.

At iteration time $j+1$, the value of node $\omega_{i,t}$ is updated using the calculated $\Delta \omega_{i,t}$ from equation (3), as expressed by $\omega_{i,t}^{j+1} \leftarrow \omega_{i,t}^j + \Delta \omega_{i,t}$. Consequently, the featured vectors move closer to the selected series $\mathbf{IN}_k$.

In this paper, we use the neural gas method to generate scenario trees. This is an artificial neural network algorithm used to extract the most representative vectors from observed vector samples. It is similar to the clustering analysis methods that classify the observed samples into several groups and identify the most representative member for each group. When applied to generating streamflow scenario trees, the streamflow scenarios can be viewed as different representative vectors that originate from historical streamflow series. Specifically, the neural gas method starts the

Equation (3) indicates that the recursive term ($\Delta\omega_{i,t}$) of a given node is jointly determined by the iteration step size and the weighted average distance from its scenarios to the selected series. Note that one node can be shared by more than one scenario. Also, a decreasing step size ensures the convergence of iterations. As the iteration proceeds, the sampled scenario tree will converge gradually to the centroid of the streamflow series. The centroid is the most centered point with the minimum overall distance to a given point set in a high-dimensional space, and the streamflow sequence vector from **IN** determines the corresponding coordinates of each point in the point set. The adaptation stops when iteration time has reached a pre-determined threshold.

4. Probability of scenario

We calculate the probability of each scenario according to the number of series that are the minimum distance from this scenario, using the following equation:

$$P(\omega_i) = \text{Count} \left\{ k' \in [1, K], k' | d_{i,k'} = \min_{i' \in [1, I]} \{d_{i',k'}\} \right\} / K, \quad (4)$$

in which $P(\omega_i)$ is the probability of scenario ω_i and $\text{Count}\{\cdot\}$ is a number counting function.

In contrast to other clustering-based scenario tree generation techniques [Casey and Sen, 2005; Heitsch and Römisch, 2009], the neural gas method uses a prespecified scenario tree structure as input and maintains the tree structure during the iteration processes. For incorporating the influences of tree structure variation on scenario tree generation as well as scenario tree reduction, we conduct several groups of repetitive and independent numerical experiments in section 2.3 and randomize the structure of the full tree that is used as a comparison benchmark in each group of experiments.

Figures 2a–2d show a test case for applying the neural gas method to extract scenario trees. As shown in Figure 2a, we generate a few random samples via adding a white noise series to given centroids. We use the randomly generated samples as the input to the neural gas method and set the tree structure the same as the original centroids. Figures 2b–2d show the scenario trees generated from the neural gas method at different iteration times. As iteration proceeds, the scenario tree gradually converges to the original centroids, showing that the neural gas method is able to restore the original centroids from the random samples.

2.2. Stochastic Programming With Recourse for Hydropower Operations

In a multistage stochastic programming with recourse model, the stages are usually divided into two parts: the immediate stage (here and now) and forthcoming stages (wait and see). When the model is applied to hydropower optimization, we use time periods to represent stages. In the immediate time period, we make a unique decision on the release policy based on deterministic streamflow forecasts. Therefore, we obtain deterministic outcomes of the reservoir ending storage and benefit associated with the release policy. Since information beyond the immediate time period is uncertain, release outcomes in the forthcoming time periods (beyond the immediate time period) are stochastic and determined by the corresponding scenarios. In real-time operation, only the deterministic release policy in the immediate time period is implemented. As time proceeds, a recourse action is taken based on the updated information in the new time period in order to mitigate possible negative consequences caused by the operation strategy in the previous time period. This process continues period-by-period with updated new information.

2.2.1. Objective Function

We use a monthly time period in our study. The objective is to maximize expected total energy production of the reservoir system during the planning horizon. The objective function can be formulated as

$$\max \text{Es}(\{\omega_i\}) = \sum_{m=1}^M E_{m,1} + \sum_{i=1}^I P(\omega_i) \cdot \sum_{m=1}^M \sum_{t=2}^T E_{m,t}^i, \quad (5)$$

in which $E_{m,t}^i$ is the energy production from reservoir m in scenario i and time period t (MWh). Multiplied by the scenario probability $P(\omega_i)$, the term $\sum_{i=1}^I P(\omega_i) \cdot \sum_{m=1}^M \sum_{t=2}^T E_{m,t}^i$ denotes the expected value of energy production from the reservoir system beyond the first time period. $\text{Es}(\{\omega_i\})$ represents the value of the

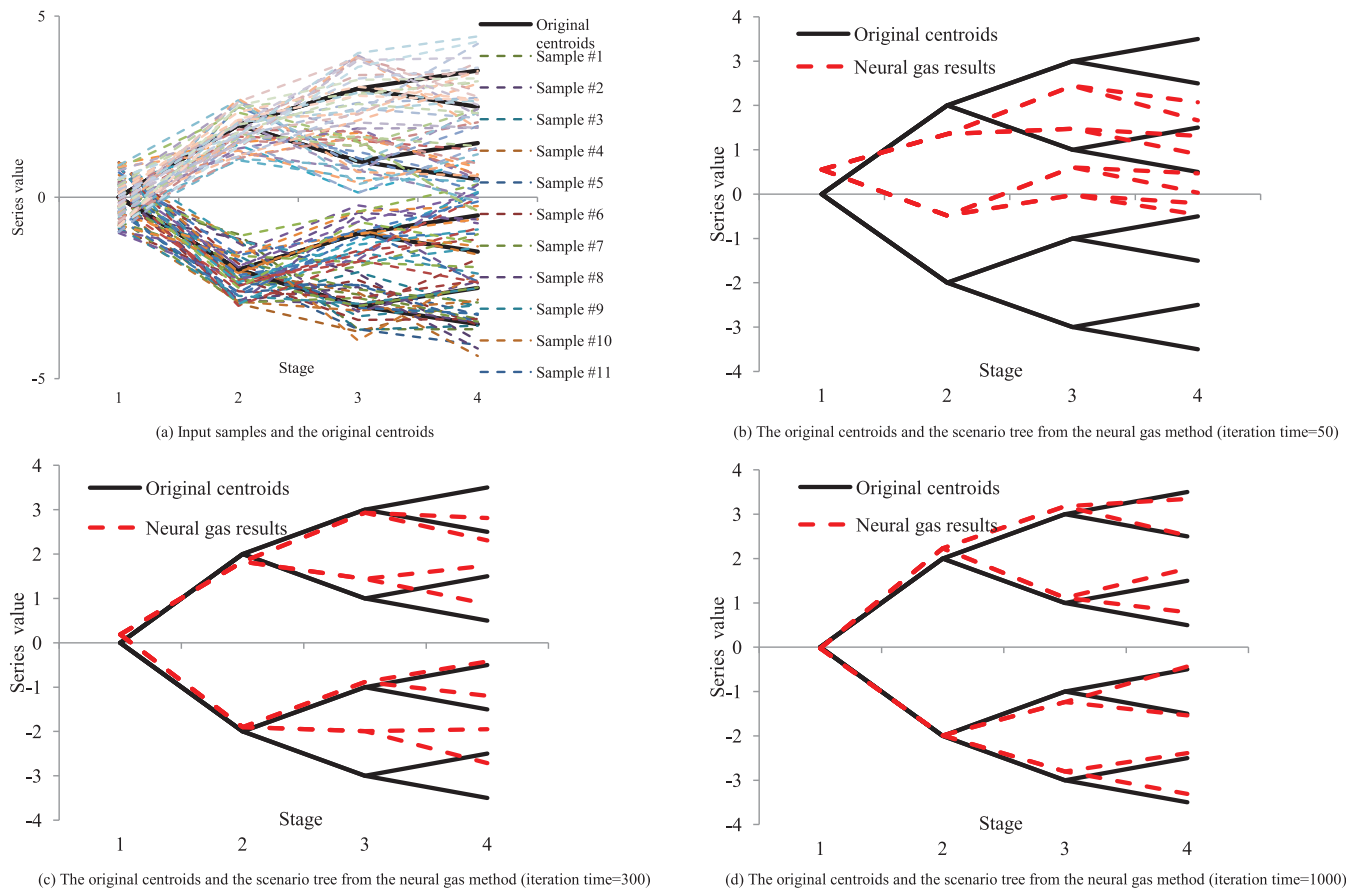


Figure 2. A test case for the neural gas method.

objective function associated with scenario tree $\{\omega_i\}$. Note that energy production in the first time period (immediate time period) is deterministic such that the term $E_{m,1}$ is used without superscript i , while power generation beyond the immediate time period ($E_{m,t}^i$) is stochastic and is associated with each specific scenario ω_i and the corresponding probability $P(\omega_i)$.

2.2.2. Constraints

1. Mass balance equation:

$$S_{m,t+1}^i = S_{m,t}^i + (W_{m,t}^i - R_{m,t}^i - SP_{m,t}^i - EV_{m,t}^i) \cdot \Delta t_t, \quad (6)$$

where $S_{m,t+1}^i$ and $S_{m,t}^i$ are the storage of reservoir m at the ending and beginning of time period t under scenario i , respectively (m^3); $W_{m,t}^i$, $R_{m,t}^i$, and $SP_{m,t}^i$ are the inflow, power release, and nonpower release (spill), respectively, from reservoir m in time period t under scenario i (m^3/s); $EV_{m,t}^i$ is the evaporation rate [Silva and Zambon, 2013] of reservoir m in time period t under scenario i (m^3/s); and Δt_t is the time interval in each time period (s).

2. Hydraulic continuity:

$$\begin{aligned} W_{m,t}^i &= \omega_{i,t,m} + R_{m-1,t}^i + SP_{m-1,t}^i, & m \geq 2 \\ W_{m,t}^i &= \omega_{i,t,m}, & m = 1, \end{aligned} \quad (7)$$

For a cascade reservoir system, we let the index be 1 for the most upstream reservoir and M for the most downstream reservoir.

Equation (7) shows that the release from the upstream reservoir becomes the inflow to the next downstream reservoir, and so on.

3. Energy production:

$$E_{m,t}^i = R_{m,t}^i \cdot \eta(H_{m,t}^i) \cdot \Delta t_t \quad (8)$$

$$H_{m,t}^i = f_1((S_{m,t}^i + S_{m,t+1}^i)/2) - f_2(R_{m,t}^i + SP_{m,t}^i), \quad (9)$$

where $H_{m,t}^i$ is the gross average water head of reservoir m during time period t under scenario i (m), which is the difference between average forebay water level and tailrace water level; f_1 and f_2 are functions that specify the forebay water level and tailrace water level, respectively; and $\eta(H_{m,t}^i)$ is the energy productivity function (MWh/m³).

4. Storage limits:

$$S_{m,t+1} \leq S_{m,t+1}^i \leq \overline{S_{m,t+1}}, \quad (10)$$

where $\overline{S_{m,t+1}}$ and $S_{m,t+1}^i$ are the upper and lower bounds, respectively, on storage at the end of time period t for reservoir m (m³).

5. Power output limits:

$$\underline{L_{m,t}} \leq E_{m,t}^i / \Delta t_t \leq \overline{L_{m,t}}, \quad (11)$$

where $\underline{L_{m,t}}$ and $\overline{L_{m,t}}$ are minimum and maximum limits of power output (MW), respectively.

6. Initial and boundary conditions:

$$S_{m,1}^i = SB_m; S_{m,T+1}^i = SE_m, \quad (12)$$

where SB_m and SE_m are the beginning storage and ending storage, respectively, of reservoir m (m³).

7. Uniqueness on decision variables

The results of decision variables (power release and nonpower release) for the same node shared by more than one scenario should be the same:

$$R_{m,t}^{i_1} = R_{m,t}^{i_2}; SP_{m,t}^{i_1} = SP_{m,t}^{i_2} \quad \text{if } \omega_{i_1,t} = \omega_{i_2,t}, \forall i_1, i_2 \in [1, I] \quad (13)$$

2.2.3. Recourse Decision

Although information beyond the immediate time period is uncertain, the uncertainty is minimized gradually as time moves forward. To correct a possibly incorrect decision made in the previous time period, a recourse action is taken in the next time period based on the updated information. To obtain the recourse strategy, we use the same model structure introduced earlier with the updated observed and forecasted information. However, the planning horizon needs to be modified. This can be achieved by either rolling the entire planning horizon [Zhao *et al.*, 2012] in a moving window or only moving the first time period while keeping the last time period fixed [Yeh, 1985]. For example, if a rolling horizon is used, we can change the planning horizon to $[2, T+1]$ and rerun the model at the end of the first time period. Alternatively, if a fixed boundary condition is used at T , at the end of the first time period we rerun the model from 2 to T ($[2, T]$).

In this study, we use the second approach, i.e., the recourse action is taken by moving the first time period (immediate time period) and running the model to the fixed boundary condition.

2.3. Numerical Experiments

As mentioned previously, scenario tree reduction will cause information loss. As a result, the reduced solution may be less accurate. Although scenario tree reduction lessens computational burden, it is at the risk of degrading the model solution. Therefore, it is extremely important to ascertain the trade-off relationship between model performance and scenario tree reduction so that an optimum level of scenario tree reduction can be determined.

In real-world applications, we can only use a limited number of scenarios to represent the original continuous stochastic process. Consequently, the accuracy of the objective function and quality of solution of the stochastic model are influenced by the scenario tree that approximates the original continuous stochastic process. To assess the accuracy of the objective function when using a reduced scenario tree, Kaut and Wallace [2007] proposed in-sample and out-of-sample stability tests.

In-sample stability states that if we generate several different scenario trees and use them as the inputs to a stochastic model, the objective function values under these trees should be nearly the same as long as the total number of scenarios from these trees is the same. This can be expressed as

$$Es(\{\omega_{l_1}\}) \approx Es(\{\omega_{l_2}\}), \quad l_1 = 1, 2, \dots, I; l_2 = 1, 2, \dots, I, \quad (14)$$

in which $\{\omega_{l_1}\}$ and $\{\omega_{l_2}\}$ are two different scenario trees with the same number of scenarios.

Unlike in-sample stability, which defines the robustness of the objective function, out-of-sample stability requires minimizing the solution bias. The bias is measured against the “true” optimal solution. We use a large reference tree to represent the “true” distribution of the original stochastic process. Specifically, out-of-sample stability implies that if we substitute the solutions obtained from different scenario trees with the same number of scenarios into the original stochastic model—as represented by its deterministic equivalent based on the large reference tree—the objective function values should be nearly the same. This can be expressed as

$$Es(\{\omega_L\}; \arg \max Es(\{\omega_{l_1}\})) \approx Es(\{\omega_L\}; \arg \max Es(\{\omega_{l_2}\})), \quad (15)$$

in which $\{\omega_L\}$, $L = 1 \dots IR$ is the reference tree and IR is the total number of scenarios from the large reference tree.

2.3.1. In-Sample and Out-Of-Sample Tests

In order to investigate the impact of scenario reduction on the objective function, we conduct in-sample and out-of-sample tests, respectively, according to the procedures suggested by Fleten *et al.* [2002].

Prior to the tests, we use the neural gas method to generate a reference tree $\{\omega_L\}$, $L = 1 \dots IR$ that represents the “true” distribution. Based on the reference tree, we then generate G groups of trees, where G is the total number of groups. In each group, there are ten trees with scenarios of IF , $0.9IF$, $\dots$, $0.2IF$, and $0.1IF$, representing different levels of scenario tree reduction, i.e., 0%, 10%, $\dots$, 80%, and 90%.

The ten trees are generated according to the following steps. First, we trim the reference tree by randomly removing some scenarios (numbering $IR - IF$) from the tree. Thus we obtain the tree that has IF number of scenarios, but the nodes in these scenarios have to be updated as the tree structure changes. We then rerun the neural gas method with the given tree structure and generate the IF -scenario tree with updated nodes. Subsequently, the $0.9IF$ -scenario tree is produced by removing 10% of the scenarios from the IF -scenario tree, and the value of nodes in the $0.9IF$ -scenario tree are readapted by the neural gas method as well. The other trees are generated in the same way.

For clarification, we define the following terms:

1. Reference tree
It is a unique tree that represents the “true” statistical distribution of streamflow, represented as $\{\omega_L\}$, $L = 1 \dots IR$.
2. Full tree and reduced tree
Both of these tree types are not unique. A full tree has IF number of scenarios. A reduced tree has fewer scenarios than the full tree, such as $0.9IF$, $\dots$, $0.2IF$, and $0.1IF$.
3. Larger tree and smaller tree
A larger tree has more scenarios than a smaller tree. The two trees can be a full tree versus a reduced tree or two reduced trees.
 1. In-sample test
We solve the stochastic model (without recourse) introduced in section 2.2 with the input of each scenario tree from groups of trees. We then validate the in-sample stability using equation (14).
 2. Out-of-sample test
As mentioned earlier, for the out-of-sample test, we should use the deterministic equivalents of the original stochastic model to assess the objective function. However, a reduced solution cannot be substituted directly into the deterministic equivalent model, since the reduced tree and the reference tree have different nodes and structures—except for the first stage, where both trees have only one node [Kaut and Wallace, 2007]. We can only substitute the first-stage solution obtained from a reduced tree into the reference tree.
We denote the optimal solution of power release and nonpower release obtained from a given reduced tree ($\{\omega_{l_1}\}$, $l_1 = 1, \dots, I$; $I < IF$) as $R_{m,t}^{l_1 \sim}$ and $SP_{m,t}^{l_1 \sim}$, respectively. The optimal solution of power release and nonpower release obtained from the reference tree are $R_{m,t}^{L \vee}$ and $SP_{m,t}^{L \vee}$, respectively.
For testing out-of-sample stability, Fleten *et al.* [2002] suggested the following steps:

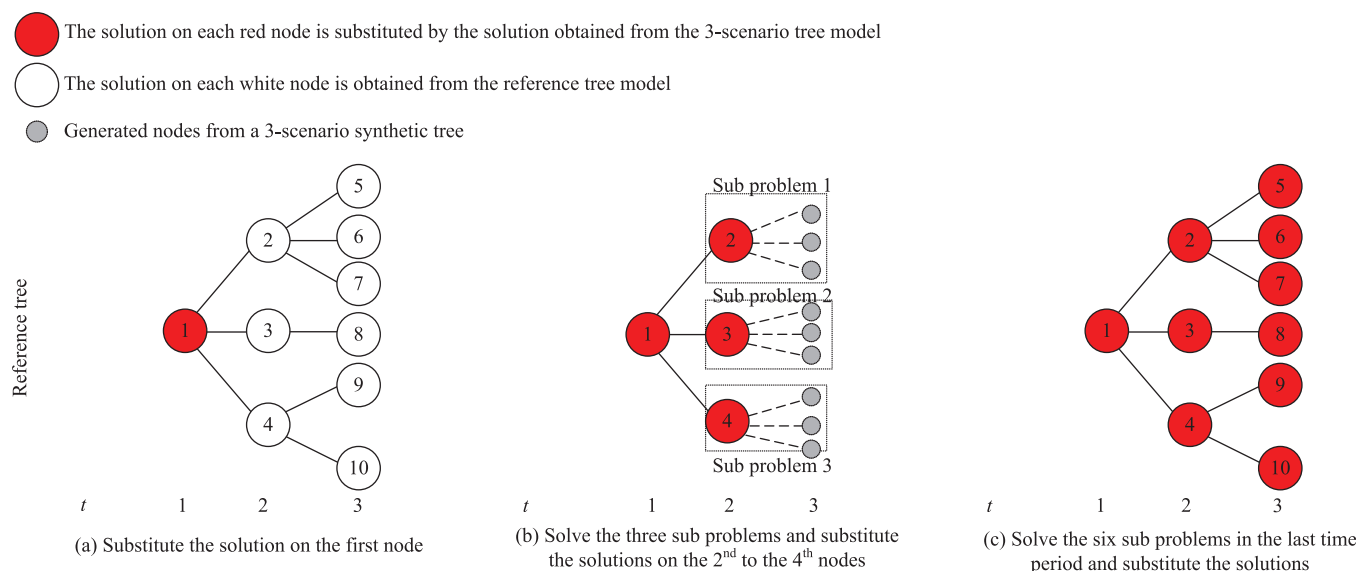


Figure 3. Schematic diagram of the out-of-sample tests.

1. Solve the stochastic programming model for a given reduced tree ($\{\omega_{l_i}\}, l_i=1, \dots, L; L < IF$) and obtain the solution for the first node in the immediate time period (refer to Figure 3).
2. Set the value of the solution on the first node in the reference tree as the same value as the solution obtained from the reduced tree, i.e., $R_{m,1}^\vee = R_{m,1}^\sim$ and $SP_{m,1}^\vee = SP_{m,1}^\sim$.
3. Let the immediate time period be $t=1$. For each scenario L ($L=1 \dots IF$) from the reference tree, do the following:
 - 3.1. Update the ending storage ($S_{m,t+1}^\vee$) at the end of the immediate time period, using equations (6) and (7) with the inputs from the unregulated streamflow in the immediate time period $\omega_{L,t,m}^\vee$, the power release $R_{m,t}^\vee$ and the nonpower release $SP_{m,t}^\vee$.
 - 3.2. Move the immediate time period to the next time period, characterized as $t \leftarrow t+1$. Generate a synthetic scenario tree that has the same number of scenarios and tree structure as the given reduced tree. The synthetic scenario tree branches-out from the current node $\omega_{L,t}$ in the reference tree, which is used for obtaining the solution on the current node when the reduced tree is "transplanted" into the reference tree.
 - 3.3. Solve the problem for the synthetic scenario tree and obtain the release solution in the immediate time period.
 - 3.4. Repeat steps (3.1) to (3.3) until the solution on the last node of scenario L ($\omega_{L,T}$) is obtained.
4. After all the solutions on the nodes in the reference tree have been "substituted" by the solutions obtained from the synthetic scenario trees, stop the procedures and evaluate the value of $Es(\{\omega_L\}; \arg \max Es(\{\omega_L\}))$ using the results on each node of the reference tree.

Figures 3a–3c show the procedures of "substituting" the solutions from a three-scenario tree to a reference tree with six scenarios. The solution for the first node (node 1) in the reference tree ($\{\omega_L\}, L=1 \dots 6$) can be substituted directly by the solution from the reduced tree ($\{\omega_{l_i}\}, l_i=1, \dots, 3$). We then update the reservoir storage for nodes 2, 3, and 4. However, the solutions for nodes 2, 3, and 4 on the reference tree cannot be substituted by the reduced tree solutions, as the two trees may have different tree structures. Instead, we generate some synthetic scenario trees that have the same tree structure and scenario numbers as the reduced tree. The synthetic scenario trees are generated based on nodes 2, 3, and 4, as described in steps 3.1 and 3.2. Note that the other nodes in each synthetic tree are readapted using the neural gas method, except the root node (the first node in the synthetic tree) from which the synthetic tree originates. With the generated synthetic scenario trees and updated initial storage for nodes 2, 3, and 4, we then obtain the solutions for these nodes by solving three subproblems based on the synthetic scenario trees (as indicated in step 3.3). We repeat the procedures until all the solutions for the nodes in the reference tree have been "replaced" with the reduced-tree solutions.

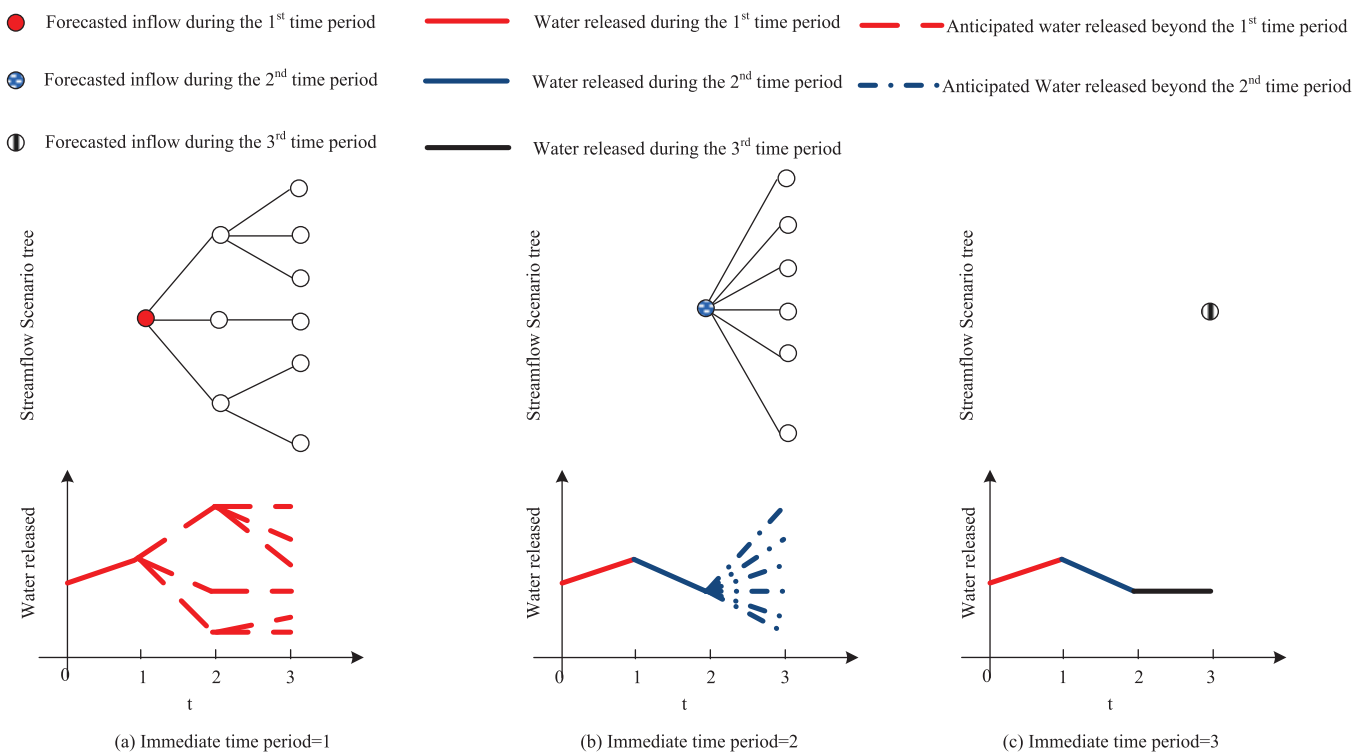


Figure 4. Schematic diagram of stochastic programming with recourse for reservoir operation.

2.3.2. Solution Quality Assessment

The in-sample and out-of-sample tests examine the stability of the objective function when scenario trees are reduced. However, these two tests cannot identify possible changes in the solutions in the immediate time period when different levels of scenario tree reduction are considered. Identifying changes in the reduced solutions is important in real-time decision making, as the release decision in the immediate time period will be implemented in a real-time operation, affecting the actual benefit.

We apply a stochastic programming with recourse model to the real-time operation of a cascade reservoir system to evaluate the quality of the reduced solutions. Specifically, we select several typical streamflow sequences (hydrology patterns) from historical records to represent the actual streamflow that will occur in real-time operations and implement the release strategies obtained from the stochastic programming with recourse model for different scenario trees. Consequently, if there are deviations among the release decisions, the energy production results from different release decisions will vary and can be used to evaluate the impact of tree reduction on solution quality.

As shown in Figure 4, we solve the corresponding stochastic programming with recourse model to determine the release strategies in each time period. As stated in section 2.2.3, we only implement the release strategy in the immediate time period and use observed streamflow in this time period to compute the ending storage. As time proceeds to the next time period, we update the storage with new information on streamflow forecast. The recourse action is executed period-by-period to the end of the selected hydrology year.

We repeat the recourse action procedure for each level of scenario reduction (from 0% to 90% with a 10% interval) and obtain the storage trajectories. We then assess the performance of the reduced solution by calculating the energy production for each storage trajectory and each selected hydrology year. The results can be used to assess the impact of scenario tree reduction on energy production in real-time operation.

In the application of a stochastic programming with recourse model, we assume a perfect streamflow forecast in the immediate time period, but streamflow beyond the immediate time period is stochastic. This

Table 1. Input Data to the Model and Reservoir Characteristics of the Cascade System of Reservoirs

Reservoirs	SB_m (10^8m^3)	SE_m (10^8m^3)	$\overline{S_{m,t+1}}$ (10^8m^3)	$S_{m,t+1}$ (10^8m^3)	$L_{m,t}$ (MW)	$\overline{L_{m,t}}$ (MW)
Shuibuya	43.12	30	43.12/41.30/38.15 ^a	19.29	104	1840
Geheyan	30.18	25	30.18/25.25	11.18	62.3	1212
Gaobazhou	4.03	4.03	4.03	3.49	20.5	270

^aThe different values of storage limits are the maximum allowable storage in the drawdown season, refill season, and flood season, respectively.

means the forecasted streamflow in the immediate time period is equal to the actual streamflow. This assumption is reasonable as periodic updates in information help improve inflow prediction in the immediate time period [Kelman *et al.*, 1990].

3. Case Study

We apply the proposed methodology to the Qingjiang cascade reservoir system, one of the important water conservation projects in China. The reservoir system is located in the middle part of the Yangtze River. There are three reservoirs in the Qingjiang cascade system, Shuibuya (the most upstream reservoir), Geheyan, and Gaobazhou (the most downstream reservoir). Shuibuya and Geheyan are storage reservoirs and Gaobazhou is a run-of-river hydropower station. This cascade reservoir system is renowned for hydropower generation. The energy produced is used to supply the demand in Hubei province and regulate load peak for the Central China Grid. The entire cascade system is owned and operated by the Hubei Qingjiang hydroelectric development corporation.

We establish a stochastic optimal operation model for this system. For each reservoir, we fit the forebay water level (a function of the storage) and the tailrace water level (a function of the total release) using second-order polynomials. The hydropower productivity function for each reservoir ($\eta(H_{m,t}^i)$) is derived from the historical records using second-order polynomial regression as well.

There are 59 years of monthly historical unregulated streamflow series from January 1951 through December 2009. In order to assess the quality of the reduced solutions (the solutions based on reduced trees) when implementing them in real-time operations, we consider three representative hydrology patterns. Specifically, we select three typical hydrology years from the historical records to represent different "actual" hydrologic patterns: wet, normal, and dry hydrology years—1967, 1979, and 1990, respectively. The other 56 yearly streamflow series are used for generating streamflow scenario trees, representing the streamflow information at hand before the actual streamflow is realized.

For each selected hydrology year, we conduct numerical experiments using procedures presented in section 2.3. Note that the recourse strategies are only implemented in the solution assessment tests, not in the in-sample and out-of-sample tests. This is because scenario tree reduction impact on the objective function (expected energy production) is greatest when the entire planning horizon is considered. Therefore, for the in-sample and out-of-sample tests, the planning horizon for each hydrology year starts on 1 January and ends on 31 December, with the immediate time period fixed in January. On the other hand, the planning horizon for solution assessment changes with shifts of the immediate time period, which starts from the immediate time period and ends on 31 December.

The reservoir characteristics and constraints are provided in Table 1.

3.1. Scenario Tree Generation and Reduction

We calibrate the parameters for the neural gas method as follows: the maximum iteration time $jm=3000$, the parameter $\lambda_0=10$ while $\lambda_f=0.01$, and the step size $\varepsilon_0=0.5$ while $\varepsilon_f=0.05$. The number of scenarios from the reference tree $IR=43$ and from the full tree $IF=33$. The total number of groups (G) is set at 10. To analyze the impact of scenario tree reduction, we generate scenario trees and reduce them using the methodology described in section 2.3.1.

Figure 5 shows the statistical moments calculated from the historical streamflow series and the reduced scenario trees for each month (using Shuibuya as the example). Figure 5a indicates that the streamflow

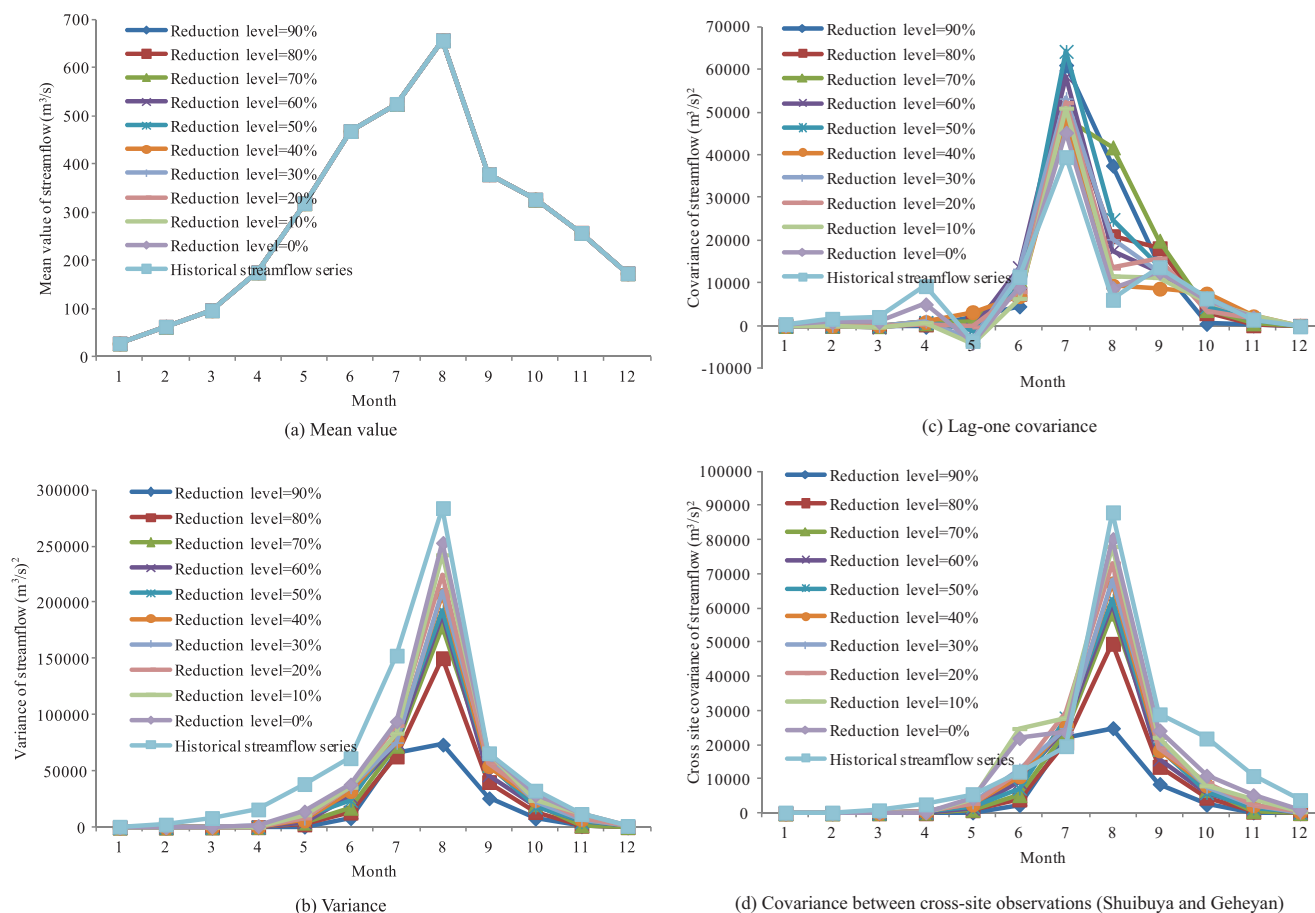


Figure 5. Comparison of the statistical moments between the historical streamflow series and the streamflow scenario trees.

scenario trees generated by the neural gas method are unbiased, as the mean values from different reduction levels remain virtually the same as the historical means. This is because the neural gas method approaches the centroids of the historical streamflow series by gradual adaptation. The mean value is an important statistical moment to preserve, as any bias introduced by a reduced tree will propagate directly to its solution.

Figures 5b, 5c, and 5d show that the variance and covariance are not preserved. The variance decreases as the reduction level increases. Note that the failure to preserve the variance of the historical streamflow series is not unique to the neural gas method. This occurs in other clustering-based methods as well, since the diversity of elements in a small cluster is generally less than that in a large cluster, given both clusters are sampled from the same original set. Figure 5c indicates that the lag-one covariance in the flood season (June–September) increases with the reduction level, while the lag-one covariance in the dry season (October–April) increases as the reduction level decreases. Figure 5d implies that the cross covariance decreases as reduction level increases.

The reduction in the streamflow variance caused by scenario tree reduction will result in streamflow diversity loss. As a result, the reduced streamflow scenario tree only contains medium-level streamflow scenarios; the extremely dry and wet scenarios are not reproduced when the tree is systematically reduced.

3.2. Impact of Scenario Tree Reduction on the Objective Function

We use the trees obtained in section 3.1 and the constraints listed in Table 1 as inputs to the stochastic model formulated in section 2.2. Note that the deterministic equivalents of the stochastic programming problem are nonlinear because of the nonlinear power productivity function. We use a nonlinear solver, LINGO (http://www.lindo.com/index.php?option=com_content&view=article&id=2&Itemid=10), to solve

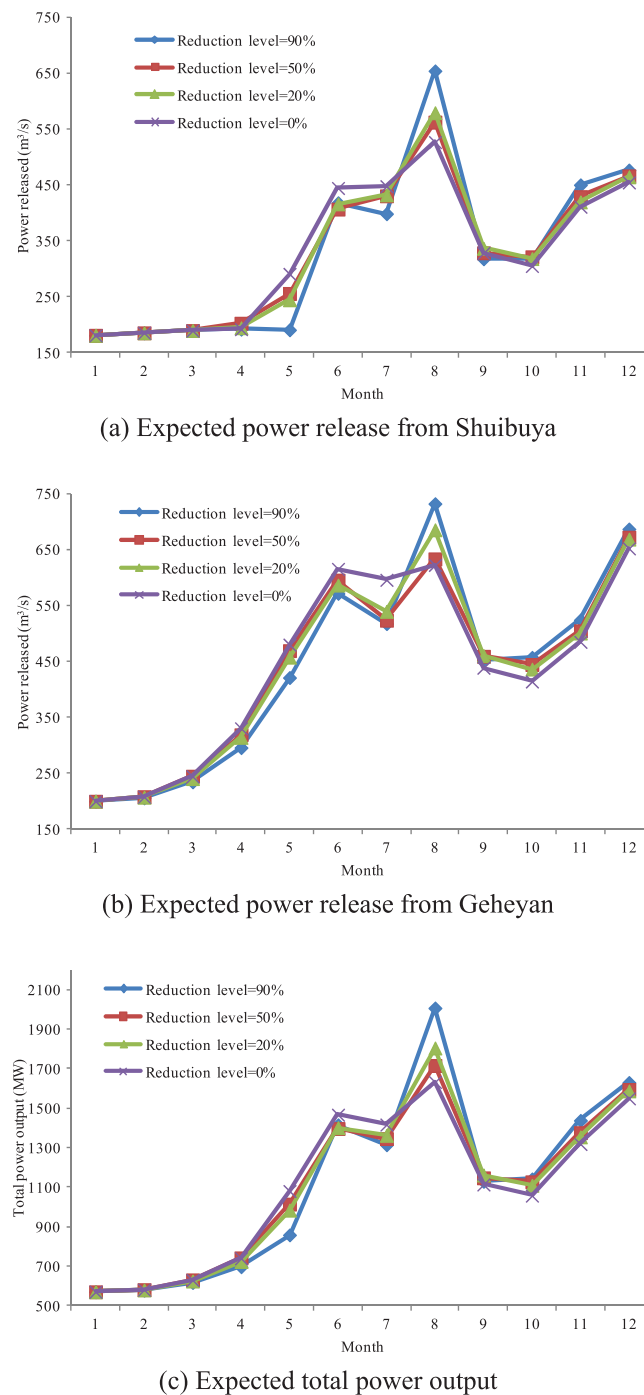


Figure 6. Expected power release and power output trajectories from different scenario trees.

tions obtained from the reduced trees are often so optimistic that they hedge the least, resulting in benefit loss in real-time operations when the actual streamflow status is dry. Therefore, we can infer from the above results that release decisions based on a larger tree are more risk neutral compared with a smaller tree, since more diverse streamflow scenarios are incorporated with a larger tree. Consequently, when using different scenario trees, the reservoir operation outcomes vary greatly, indicating that solution accuracy deteriorates due to scenario tree reduction.

the deterministic equivalents. For each month in the normal hydrology year (1979), Figures 6a and 6b show the expected value of power release from Shuibuya and Geheyan, respectively, while Figure 6c shows the expected value of total power output from the cascade reservoir system when different streamflow scenario trees are used as inputs.

Figures 6a and 6b indicate that, from January to March, the expected values of power release strategies on different trees are quite similar. From April to July, the power release obtained from a larger tree is greater than that from a smaller tree. The increased volume of power release before the flood season (from June to September) is the prerelease, which helps reduce spillage before the occurrence of flood events under extremely wet scenarios. In August, the power release increases with the reduction level, due to the fact that nonpower release decreases as reduction level increases. This is because more wet and dry scenarios are sampled in a larger tree (a scenario tree that has more scenarios) and more spillage occurs under those wet scenarios compared with the result from a smaller tree (a scenario tree that has fewer scenarios). Note that the power release after the flood season (from October to December) increases with the reduction level, which is different from the release policy before the flood season. The reason is that the reduced volume of power release obtained for a larger tree helps mitigate future energy shortages in case those extremely dry scenarios are anticipated after the flood season.

As scenario tree reduction decreases the diversity of the scenarios, extreme wet and dry scenarios are excluded in a reduced tree. As a result, the solu-

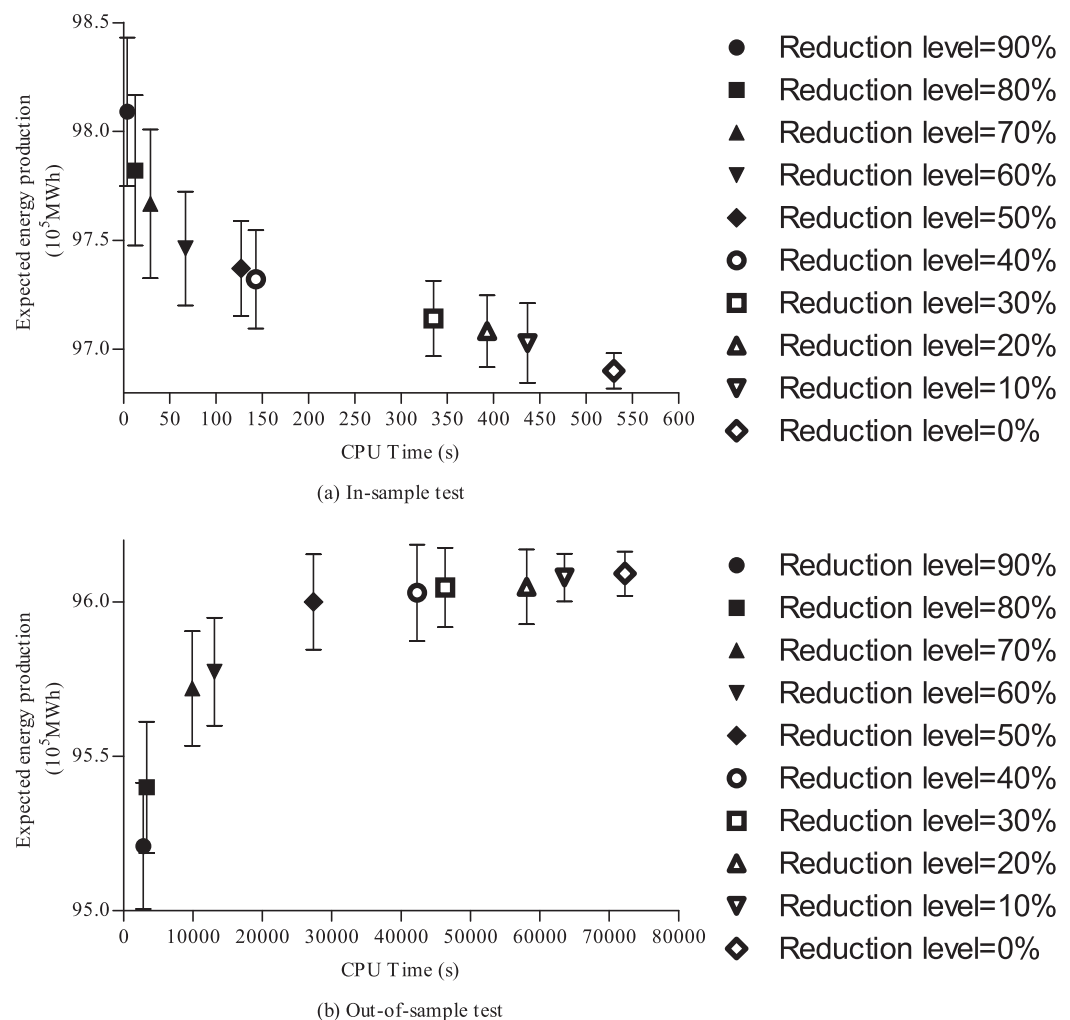


Figure 7. Trade-off between objective function and CPU time in the in-sample and out-of-sample tests.

Figure 6c shows that the expected value of power output increases with the reduction level. The major difference appears at the flood season, during which spillage may occur.

To evaluate the impact of scenario tree reduction on the objective function, we conduct the in-sample and out-of-sample tests as described in section 2.3.1. The trade-off between expected energy production (objective function) and CPU time is plotted in Figure 7. Note that the CPU times plotted in Figure 7 are the statistical results from the G groups of tests. As Figure 7 indicates, the mean values of expected energy production from the in-sample and out-of-sample tests reveal different trends. Results from the in-sample test, shown in Figure 7a, indicate the mean values of expected energy production decrease as CPU time increases. The results also show that the mean value of expected energy production decreases as the reduction level decreases. This is because larger trees include more wet inflow scenarios than smaller trees. This causes an increase in the expected spillage and, consequently, reduces expected energy production. The in-sample test results show that the reduced model overestimates model performance in terms of expected energy production.

In contrast, the results from the out-of-sample tests show different trends. Figure 7b indicates that the mean value of expected energy production increases as CPU time increases, and the mean value of expected energy production increases as the reduction level decreases. The reason is that the gap between a reduced solution and the “true” optimal solution narrows as the scenarios increase. Consequently, the objective function value improves.

Table 2. Dimensionality of the Deterministic Equivalents of the Stochastic Programming Models (Without Recourse)

Reduction Level	Nonlinear Variables	Total Variables	Nonlinear Constraints	Total Constraints	Iteration Numbers	Memories (kb)	CPU Time (s)	
							Out-of-Sample	In-Sample
0%	4,020	9,175	2,412	7,965	21,660	1,930	72,325	530
10%	3,465	7,909	2,079	6,862	13,081	1,620	63,577	437
20%	3,045	6,949	1,827	6,032	11,503	1,444	58,105	393
30%	2,655	6,059	1,593	5,260	11,209	1,270	46,381	335
40%	2,370	5,407	1,422	4,691	9,654	1,149	42,312	143
50%	2,250	5,133	1,350	4,453	8,229	1,099	27,411	127
60%	1,845	4,207	1,107	3,646	6,863	880	13,142	67
70%	1,590	3,623	954	3,135	5,611	772	9,955	29
80%	1,050	2,391	630	2,067	3,932	528	3,378	13
90%	555	1,263	333	1,092	2,225	304	2,883	4

In both in-sample and out-of-sample tests, the variation range of expected energy production increases as the reduction level increases, showing that scenario reduction also affects the model solution's robustness.

Table 2 lists the dimensionality of the deterministic equivalents of the stochastic programming models (without recourse) for different trees. It is evident that the dimensionality grows drastically with an increase in scenarios. The CPU times required for solving the full-tree model are 72,325 s and 530 s for the out-of-sample and in-sample tests, respectively. With a 90% reduction level, the CPU times decreased to 2883 s and 4 s, respectively. This clearly shows the trade-off between the level of scenario tree reduction and CPU time. The task now is to determine the optimum level of scenario tree reduction such that the bias in the objective function value is acceptable.

To achieve this, we use Welch's t test [Welch, 1947] to examine whether the mean value of the objective function from a reduced solution is significantly different from the mean value of the full-tree solution at a given significance level. The null hypothesis HP_0 is that the mean value of the objective function from the reduced tree solutions is equal to the mean value of the objective function from the full-tree solutions:

$$HP_0 : \frac{1}{G} \sum_{g=1}^G Es(\{\omega_{lg}\}) = \frac{1}{G} \sum_{g'=1}^G Es(\{\omega_{lg'}\}), l_g = 1, \dots, l; l < l_F; l_g = 1, \dots, l_F \quad (16)$$

We formulate the t-test variable ($ttest_1$) for the null test as follows:

$$ttest_1 = 1/G \cdot \left[\sum_{g=1}^G Es(\{\omega_{lg}\}) - \sum_{g'=1}^G Es(\{\omega_{lg'}\}) \right] / \sqrt{\sigma_1^2/G + \sigma_2^2/G} \quad (17)$$

where σ_1^2 and σ_2^2 are, respectively, the variance of the objective function value from the reduced tree solutions and the variance of the objective function value from the full-tree solutions. The variable ($ttest_1$) obeys the t-distribution that has a degree of freedom of $v_1 \approx (\sigma_1^2/G + \sigma_2^2/G)^2 / \{[\sigma_1^4 + \sigma_2^4]/[G^2(G-1)]\}$.

Similarly, the null hypothesis for the out-of-sample results is that the mean value of the objective function under the out-of-sample criterion from the reduced tree solutions is equal to the mean value of the objective function under the out-of-sample criterion from the full-tree solutions. This can be characterized as

$$HP_0 : \frac{1}{G} \sum_{g=1}^G Es(\{\omega_L\}; \arg \max Es(\{\omega_{lg}\})) = \frac{1}{G} \sum_{g'=1}^G Es(\{\omega_L\}; \arg \max Es(\{\omega_{lg'}\})), \quad l_g = 1, \dots, l; l < l_F; l_g = 1, \dots, l_F; L = 1, \dots, l_F \quad (18)$$

The t-test variable ($ttest_2$) for the out-of-sample results is

$$ttest_2 = 1/G \cdot \left[\sum_{g=1}^G Es(\{\omega_L\}; \arg \max Es(\{\omega_{lg}\})) - \sum_{g'=1}^G Es(\{\omega_L\}; \arg \max Es(\{\omega_{lg'}\})) \right] / \sqrt{\sigma_1'^2/G + \sigma_2'^2/G} \quad (19)$$

in which $\sigma_1'^2$ and $\sigma_2'^2$ are, respectively, the variance of the objective function value under the out-of-sample criterion from the reduced tree solutions and the variance of the objective function value from the full-tree

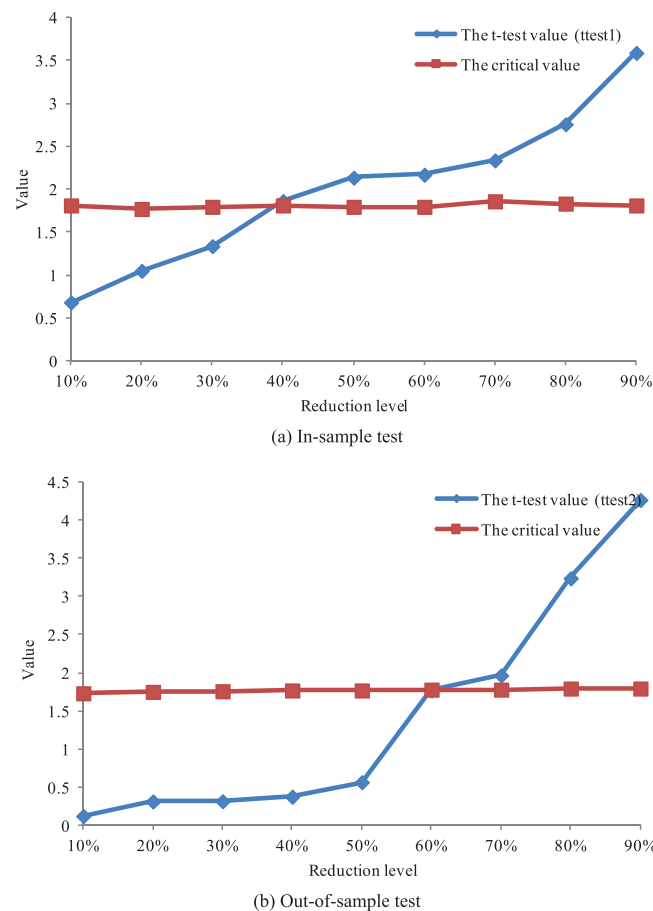


Figure 8. The results from Welch's t test.

both the in-sample and out-of-sample stability criteria, the highest level of scenario tree reduction is 40%. At this reduction level, expected energy production from the reduced solution in the in-sample tests increases by 0.4% compared with expected energy production from the full-tree solution. Meanwhile, expected energy production from the reduced solution in the out-of-sample tests decreases by 0.07% compared with the full-tree solution. The changes are statistically insignificant. On the other hand, the CPU time decreases from 530 s (full-tree) to 143 s (at a reduction level of 40%) in the in-sample tests while the CPU time decreases from 72,325 s to 42,312 s in the out-of-sample tests. The reduction in computational demand is significant.

3.3. Impact of Scenario Tree Reduction on Solution of the Stochastic Programming With Recourse Model

We now analyze the impact of scenario tree reduction on the solution of the stochastic programming with recourse model. We implement different reduced solutions (release decisions) with recourse, period-to-period, in the selected hydrology years, and observe the results of energy production from each reduced solution. Note that energy production (with recourse) assessed in this section is different from the expected energy production (without recourse) assessed in section 3.2. The energy production ($\sum_{m=1}^M \sum_{t=1}^T E_{m,t}$) assessed here represents the "actual" energy production that is produced in each selected hydrology year, as we have already implemented the release decisions with recourse in each time period. On the other hand, the expected energy production ($\sum_{m=1}^M E_{m,1} + \sum_{i=1}^I P(\omega_i) \cdot \sum_{m=1}^M \sum_{t=2}^T E_{m,t}^i$) is the future unknown energy estimated in the immediate time period. The storage trajectories of Shuibuya and Geheyan are plotted in Figure 9.

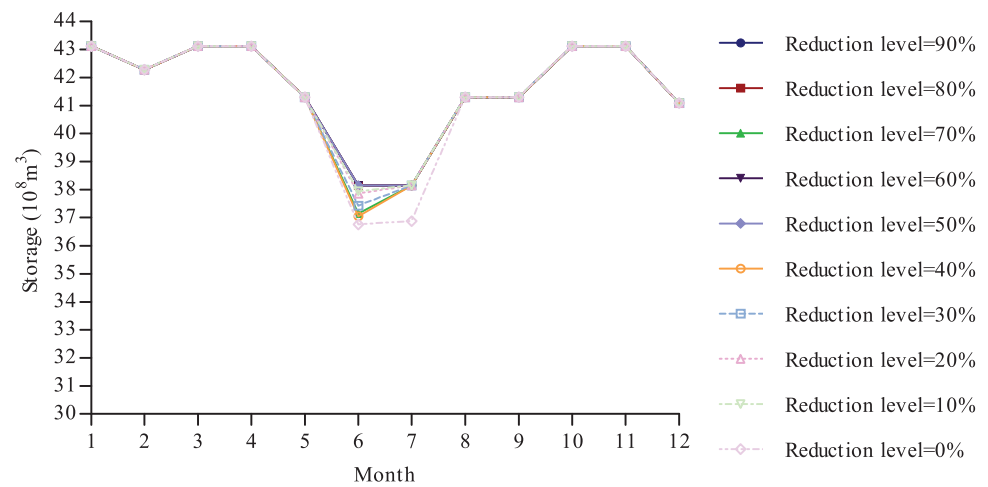
In general, in any one of the three selected hydrology years, the average storage in Shuibuya in the draw-down and flood seasons obtained from a larger tree is lower than that obtained from a smaller tree. After

solutions. The variable ($ttest_2$) obeys the t-distribution that has a degree of freedom of $\nu_2 \approx (\sigma_1'^2/G + \sigma_2'^2/G)^2 / \{[\sigma_1'^4 + \sigma_2'^4] / [G^2(G-1)]\}$.

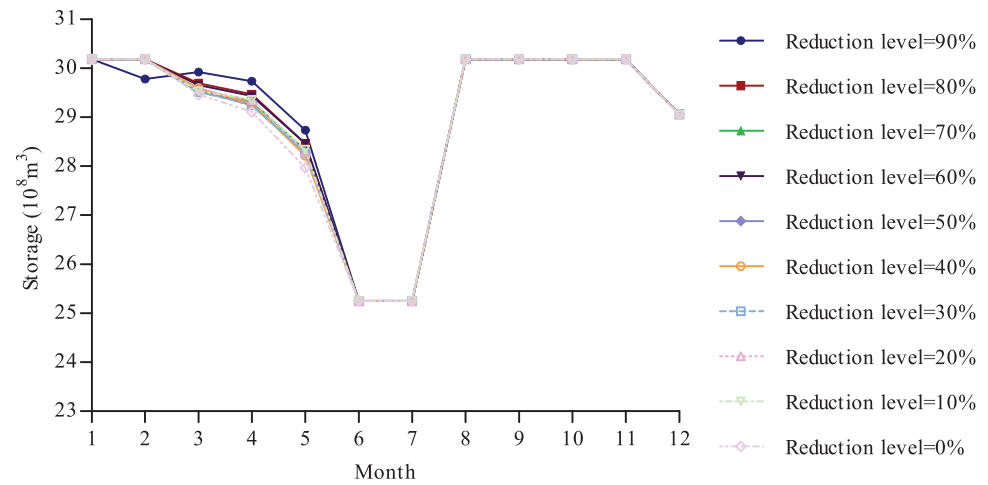
We conduct Welch's one-tailed t test on both the in-sample test results and the out-of-sample test results under a significance level of 0.05. For instance, for the in-sample test results, we compare the objective function at each scenario tree reduction level with the objective function obtained from the full trees. If the t-test value is greater than the critical value of the t distribution for the given significance level, we have the confidence to reject the null hypothesis. Otherwise, we accept the null hypothesis.

We plot the t-test value at each reduction level and the corresponding critical value in Figure 8. From the in-sample test results, the critical value intercepts the t-test value at the 40% scenario tree reduction level, whereas with the out-of-sample test results the critical value intercepts the t-test value at the 60% scenario tree reduction level.

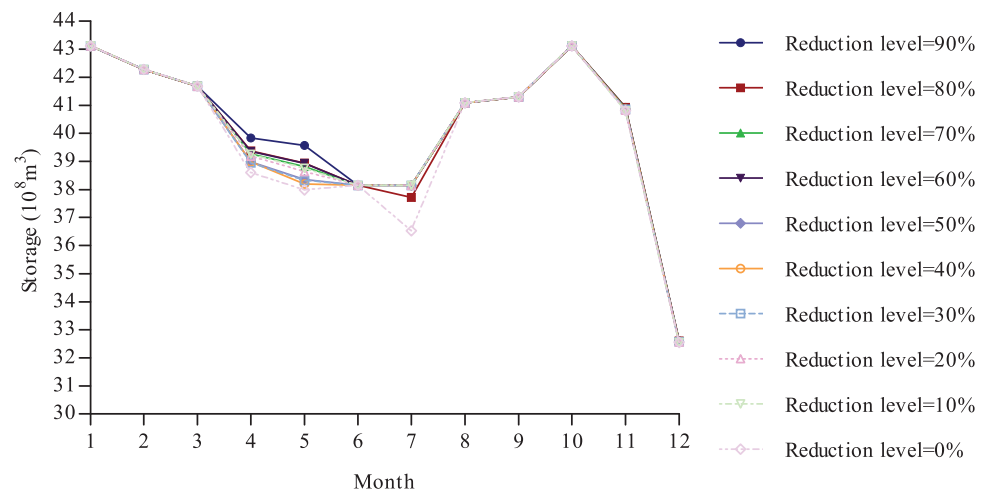
Therefore, in terms of preserving the objective function evaluated under



(a) Storage variation in Shuibuya during the wet hydrology year

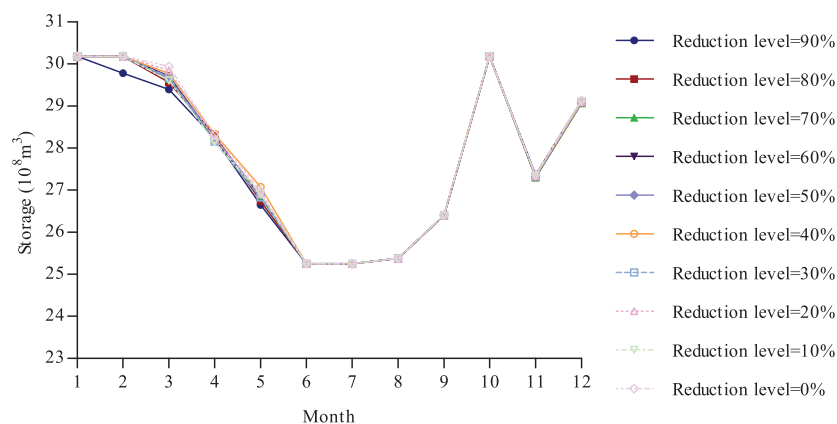


(b) Storage variation in Geheyan during the wet hydrology year

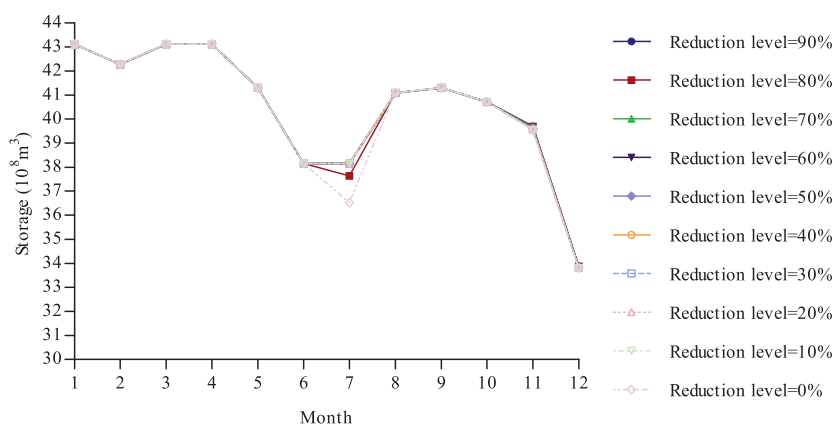


(c) Storage variation in Shuibuya during the normal hydrology year

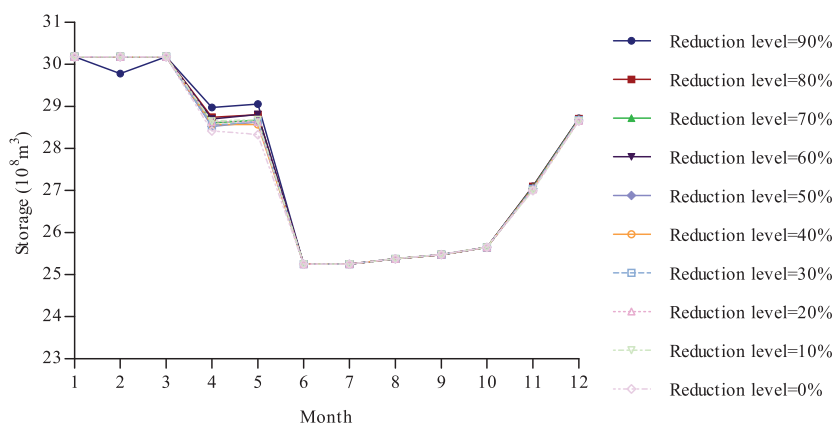
Figure 9. Storage variations in Shuibuya and Geheyan from different reduced solutions and hydrology years.



(d) Storage variation in Geheyan during the normal hydrology year



(e) Storage variation in Shuibuya during the dry hydrology year



(f) Storage variation in Geheyan during the dry hydrology year

Figure 9. (continued)

the flood season, the storages in Shuibuya obtained from different trees are almost the same. During the drawdown season, the lower storage in Shuibuya from a larger-tree solution is attributed to an increased release, which is used to reduce expected spillage in the future.

In the drawdown and flood seasons of both the wet and dry hydrology years, similar to the storage variations in Shuibuya, the average storage in Geheyan decreases as scenario numbers increase. However, the ending storage in Geheyan in January obtained from the model that uses the fewest scenarios (reduction level of 90%) deviates from other solutions significantly due to the highest level of scenario reduction. In

the normal hydrology year, the model using a larger tree as its input produces higher storage in Geheyan than the storage result obtained from a smaller tree.

Accordingly, information loss caused by tree reduction can result in different operating strategies and eventually affect the overall benefit. Table 3 presents the dimensionality and CPU time of the deterministic equivalents of stochastic programming with recourse models. Note that we sum-up these indices in each recourse step, i.e., the CPU time is the total CPU time that elapses to obtain all the recourse strategies from the first time period (January) to the last time period (December). Table 4 lists the energy production of the multireservoir system for each selected hydrology year. The statistical results under each reduction level are the average results that are calculated based on G groups of independent experiments.

We can infer from the results that:

1. Scenario tree reduction can reduce the average energy production of the system in the three selected hydrology years. It reduces energy production from 105.66×10^5 MWh to 105.4×10^5 MWh, which means that solution quality may deteriorate due to scenario tree reduction.
2. Considering all the hydrology years, the solution obtained from a larger tree does not always outperform the solution from a smaller tree. In this case study, a larger-tree solution outperforms a smaller-tree solution in only the wet and normal hydrology years, but not in the dry year. Energy production in the wet hydrology year increases from 138.14×10^5 MWh to 138.79×10^5 MWh with increasing scenarios, while energy production in the normal hydrology year changes from 95.12×10^5 MWh to 95.45×10^5 MWh. In contrast, energy production in the dry hydrology year falls from 82.93×10^5 MWh to 82.73×10^5 MWh as scenarios increase.

This outcome can be explained according to the mechanisms of stochastic optimization. That is, the optimal stochastic solution that maximizes the expected value of energy production does not guarantee that the result of energy production under each particular scenario is maximized as well. Basically, water release obtained from the stochastic model is risk-neutral; that is, it should be neither too large for preventing water/energy shortage in a future drought, nor too small for incurring water/energy loss due to future flood events. As analyzed in section 3.1, scenario tree reduction does not preserve high-order moments, such as the variance. Consequently, extreme hydrology scenarios (dry and wet scenarios) will be excluded from a reduced tree. Moreover, the volume of water released from a larger-tree solution can be lower in the beginning of the drawdown season, in which the potential impact of droughts dominates the impact of floods (as in the case of water released from Geheyan in January in the normal hydrology year). The reduced release volume helps produce a higher head and productivity at Geheyan at the beginning of the drawdown season. In contrast, the volume of water released from a larger-tree solution can be higher when the immediate time period approaches the flood season, in which the potential impact of flood dominates the drought (such as water released from Shuibuya in May of the wet hydrology year). Consequently, a larger-tree solution yields less spillage than a smaller-tree solution in wet and normal hydrology years. Both the improved productivity in the drawdown season and the reduced spillage in the flood season enhance energy production; thus energy production in the wet and normal hydrology years increases with scenarios.

There is an exception in the dry hydrology year, in which a smaller-tree solution outperforms a larger-tree solution in terms of energy production. This can occur because more prerelease in the drawdown season from a larger-tree solution can negatively affect energy production when the actual inflow in the future turns-out to be dry.

We apply Welch's t test to judge whether significant changes occur in energy production when different levels of scenario tree reduction are implemented. The results show that if the reduction level is less than 70%, there is no significant change in solution quality.

As we investigated in section 3.2, to meet the in-sample criterion, the highest level of scenario tree reduction is 40%. Moreover, for preserving the objective function value under the out-of-sample criterion, the highest level of scenario tree reduction is 60%. Therefore, for preserving the objective function values under both criteria as well as maintaining solution quality, the highest level of scenario tree reduction is 40%, which is the minimum of (40%, 60%, and 70%). At this reduction level, the computational demand of the stochastic programming with recourse model can be reduced by 72.3% in terms of CPU time (from 2585 s to 726 s).

Table 3. Dimensionality of the Deterministic Equivalents of the Stochastic Programming With Recourse Models

Reduction Level	Nonlinear Variables	Total Variables	Nonlinear Constraints	Total Constraints	Iteration Numbers	Memories (kb)	CPU Time
0%	26,130	59,984	15,678	52,834	132,929	13,033	2,585
10%	22,440	51,509	13,464	45,340	91,842	11,014	2,135
20%	19,820	45,489	11,892	40,059	81,393	9,892	1,913
30%	17,340	39,796	10,404	35,048	75,181	8,669	1,624
40%	15,350	35,213	9,210	30,985	63,099	7,804	716
50%	14,570	33,421	8,742	29,405	59,485	7,468	635
60%	11,800	27,047	7,080	23,761	48,100	5,984	347
70%	9,950	22,781	5,970	19,969	38,329	5,135	163
80%	6,550	14,979	3,930	13,107	26,673	3,571	78
90%	3,580	8,175	2,148	7,149	17,355	2,233	32

4. Discussions and Conclusions

In this paper, we established a basic stochastic programming model for optimizing total energy production from a system of cascade reservoirs. We applied the neural gas method to generate streamflow scenario trees. To determine the impact of scenario tree reduction on the results of the objective function and solution, we used Monte Carlo sampling to generate reduced trees and conducted numerical experiments on the Qingjiang system of cascade reservoirs. We determined the trade-off between the objective function and CPU time. We used a statistical hypothesis test to determine the acceptable level of scenario tree reduction. In addition, we selected three typical hydrology years and implemented a stochastic programming with recourse model with different reduced trees. We observed the changes in energy production and evaluated the impact of scenario tree reduction on solution quality.

The results show that: (1) the neural gas method produces unbiased scenario trees but does not preserve the variances and covariances when scenario trees are reduced. Consequently, the diversity of streamflow scenarios deteriorates as the level of scenario tree reduction increases. (2) Due to the diversity loss, extreme dry and wet scenarios are not reproduced in the reduced trees. As a result, both values of the objective function in the in-sample and out-of-sample tests change when different levels of scenario tree reduction are applied. (3) For preserving the values of the objective function under both the in-sample and out-of-sample criteria as well as maintaining solution quality, the highest scenario tree reduction level is 40%. (4) Under the 40% reduction level, the result of the objective function in the in-sample tests produces 0.4% more expected energy than the result from the full-tree solution, while the objective function in the out-of-sample tests produces 0.07% less expected energy. The results obtained from the stochastic programming with recourse model show that the mean value of energy production (the average of the three selected hydrology years) from the reduced solutions (at a reduction level of 40%) is 0.04% lower than the full-tree solutions. These changes are statistically insignificant. At the 40% scenario reduction level, the CPU time in the in-sample tests (without recourse) is reduced from 530 s to 143 s while the CPU time in the out-of-sample tests (without recourse) is decreased from 72,325 s to 42,312 s. Additionally, the CPU time required for solving the stochastic programming with recourse model reduced from 2,585 s to 726 s.

Table 4. Energy Production^a of the System in Different Hydrology Years

Reduction Level	Wet	Normal	Dry	Average	Standard Deviation
90%	138.14	95.12	82.93	105.40	0.17
80%	138.28	95.20	82.88	105.45	0.18
70%	138.31	95.35	82.87	105.51	0.16
60%	138.50	95.44	82.83	105.59	0.18
50%	138.55	95.44	82.84	105.61	0.18
40%	138.57	95.44	82.84	105.62	0.17
30%	138.61	95.44	82.84	105.63	0.15
20%	138.65	95.44	82.81	105.64	0.13
10%	138.69	95.44	82.79	105.64	0.13
0%	138.79	95.45	82.73	105.66	0.11

^aNote that the unit of energy production is 10^5 MWh. The standard deviation is obtained from the statistical results of the G groups of independent experiments.

The results of the studied system show that, after taking away 40% of the streamflow scenarios from the full tree, the reduced tree is still adequate to characterize the streamflow information as well as preserve solution accuracy. Therefore, decision makers can use the reduced tree to make release decisions for real-time operations. Note that the maximum reduction level is not only determined by the trade-off results between the performance indicators (the objective function value under the two criteria and the "actual" energy production obtained in real-time operations) and the reduction level, but it is also decided by the size of the full tree. If a very large tree with thousands of scenarios is used as a benchmark, the results may be different as more diverse streamflow scenarios are introduced. In such a case, if the maximum reduction level for maintaining all the performance indicators is less than 40% or much less than the decision makers' expectation, an option is to disregard the less important performance indicators (such as the objective function value under the in-sample criterion) while observing the more important indicators (such as the solution quality). This will increase the reduction level.

Information loss due to scenario tree reduction is unavoidable. The neural gas method preserves the mean value of the original streamflow series but introduces bias to the variance, cross variance, and lag-one covariance when the scenario tree is systematically reduced. This causes loss in streamflow diversity and possible serial correlation. This problem is not unique to the neural gas method, as all clustering methods and the moment matching methods suffer from the same shortcoming. Specifically, moment matching methods may fail to match the statistical moments of the observed samples perfectly when the scenario tree is reduced, resulting in information loss as well. There is room for future research in scenario tree generation.

Acknowledgments

This study is supported by the National Natural Science Foundation of China (grant 51179044), the National Basic Research Program of China (973 Program, grant 2013CB036406), the Special Fund for Public Welfare Industry of the Ministry of Water Resources of China (grant 201501007), and the São Paulo Research Foundation FAPESP in Brazil (grant 13/03432-9). The first author is supported by a fellowship from the Chinese government for his visit to the University of California, Los Angeles. Partial support also is provided from an AECOM endowment. We would like to thank three anonymous reviewers for their in-depth reviews and constructive comments. The remarks and summary of reviewer comments provided by the editor and associate editor also are greatly appreciated. The users can access the data used in this paper by contacting the first author.

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Erratum

A step on p. 6362 has been corrected, and a sentence on p. 6363 has been deleted. The authors note that the convergence criterion used for the neural gas method requires sorting scenarios in ascending distance orders in terms of the Euclidean distance between a scenario and the randomly selected series. This gives a larger weight to a closer scenario to the selected series. The adaptation step of the neural gas method can be interpreted as gradient descending on a cost function (https://en.wikipedia.org/wiki/Neural_gas), which adapts the scenarios with a step size that decreases with increasing distance in order to ensure convergence. However, our software uses ascending order rather than descending order. Consequently, there are no changes in the results of the numerical experiments or in the conclusions presented in the paper. The authors would like to thank Dr. Sean Turner of the Department of Engineering Systems and Design, Singapore University of Technology and Design, for pointing out the incorrect description to our attention. This version may be considered the authoritative version of record.